$$A(t) = U_t^\dagger A(0)\, U_t \qquad U_t = e^{-iHt/\hbar}$$

$$\frac{1}{T}\int_0^T U_t^\dagger A(t)\, U_t \, dt$$

$$f : \mathbb{R} \to \mathbb{R}$$
$$t \mapsto f(t)$$

$$\lim \frac{1}{T}\int_0^T f(t)\, dt = \text{number}$$

$A(t)$ → should think of a map $\mathbb{R} \mapsto \mathrm{Lin}(\mathcal{H}, \mathcal{H})$

<u>finite-dim case:</u> $(\mathcal{H} = \mathbb{C}^n)$ $\quad H^\dagger = H \Rightarrow \exists\, S \in U(n)$ s.t. $H = S^\dagger \overbrace{\begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}}^{D} S$

$$e^{(A^{-1}BA)} = \mathbb{1} + A^{-1}BA + \frac{1}{2}(A^{-1}BA)^2 + \dots$$
$$= \mathbb{1} + A^{-1}BA + \frac{1}{2}A^{-1}B^2A + \dots = A^{-1}e^{B}A$$

$$U_t = e^{-iHt} = S^\dagger e^{-iDt} S$$

$$\frac{1}{T}\int_0^T dt\; U_t^\dagger A\, U_t = S^\dagger \left\{ \frac{1}{T}\int_0^T e^{+iDt}\, S A S^\dagger\, e^{-iDt} \right\} S$$

$$e^{-iDt} = \begin{pmatrix} e^{-i\lambda_1 t} & & & \\ & e^{-i\lambda_2 t} & & \\ & & \ddots & \\ & & & e^{-i\lambda_n t} \end{pmatrix}$$

<u>ex: 2×2</u>

$$\begin{pmatrix} e^{i\lambda_1 t} & 0 \\ 0 & e^{i\lambda_2 t} \end{pmatrix} \overbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}^{SAS^\dagger} \begin{pmatrix} e^{-i\lambda_1 t} & 0 \\ 0 & e^{-i\lambda_2 t} \end{pmatrix}$$

$$= \begin{pmatrix} e^{i\lambda_1 t} & 0 \\ 0 & e^{i\lambda_2 t} \end{pmatrix} \begin{pmatrix} a e^{-i\lambda_1 t} & b e^{-i\lambda_2 t} \\ c e^{-i\lambda_1 t} & d e^{-i\lambda_2 t} \end{pmatrix}$$

$$= \begin{pmatrix} a & b e^{i(\lambda_1 - \lambda_2)t} \\ c e^{i(\lambda_1 - \lambda_2)t} & d \end{pmatrix}$$

<u>case 1:</u> $\lambda_1 = \lambda_2 \rightarrow \frac{1}{T}\int_0^T dt\; e^{iDt} S A S^\dagger e^{-iDt} = \frac{1}{T}\int_0^T dt\; S A S^\dagger$

$$= S A S^\dagger$$

$$\frac{1}{T}\int_0^T U_t^\dagger A U_t = A \qquad \text{if } \lambda_1 = \lambda_2 \quad H = \lambda_1 \mathbb{1} \quad \text{so } [A, H] = 0$$

<u>case 2</u> $\quad \lambda_1 \neq \lambda_2 \qquad \omega = \lambda_1 - \lambda_2$

$$S\langle A\rangle S^\dagger = \lim \frac{1}{T}\int_0^T dt\, e^{iDt} S A S^\dagger e^{-iDt} = \lim \frac{1}{T}\int_0^T dt \begin{pmatrix} a & b e^{i\omega t} \\ c e^{i\omega t} & d \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$$

$$\frac{1}{T}\int_0^T dt\, a = a$$

$$\frac{1}{T}\int_0^T dt\, b e^{i\omega t} = b \frac{e^{i\omega T} - 1}{T i\omega} \to 0$$

$$0 = [S\langle A\rangle S^\dagger, D] = [S\langle A\rangle S^\dagger, S H S^\dagger] = S[\langle A\rangle, H] S^\dagger$$

$$\Rightarrow [\langle A\rangle, H] = 0$$

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guess: infinite dimensions need H compact generator

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orthogonal projection

$$\lim_{t\to\infty} \frac{1}{T}\int_0^T dt\, U_t |\psi\rangle = \Pi |\psi\rangle$$

$$= \sum_{\substack{i\in I \\ E_i = 0}} |i\rangle\langle i|\psi\rangle$$

$\{|i\rangle \mid i \in I\}$ orthonormal basis

$H|i\rangle = E_i |i\rangle$

$$\mathcal{H}^U = \text{span}\{|i\rangle \mid i \in I,\ E_i = 0\}$$

$G = SO(3) \qquad \mathfrak{g} = \underline{so}(3) = \{X \in M_3(\mathbb{R}) \mid X^t = -X\} \to \dim 3$

$$= \left\{ \begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix} \middle| \ a, b, c \in \mathbb{R} \right\}$$

generators: a basis $\quad Y_z = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad Y_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \quad Y_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$

↳ generates rotations around z-axis

$$\exp\left[\begin{pmatrix} \boxed{\begin{matrix} 0 & 1 \\ -1 & 0 \end{matrix}} & \begin{matrix} 0 \\ 0 \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & \boxed{0} \end{pmatrix}\theta\right] = \begin{pmatrix} \exp\left[\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\theta\right] & \begin{matrix} 0 \\ 0 \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\exp\begin{pmatrix} 0 & \theta \\ -\theta & 0 \end{pmatrix} = ? \qquad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^2 = -\mathbb{1} \qquad \Longleftrightarrow \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \equiv a + ib$$

$$\begin{cases} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^{2k} = (-1)^k \mathbb{1} \\ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^{2k+1} = (-1)^k \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \end{cases}$$

$$\exp\begin{pmatrix} 0 & \theta \\ -\theta & 0 \end{pmatrix} = \sum_{n=0}^{\infty} \frac{\theta^n}{n!} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^n = \sum_{k=0}^{\infty} \frac{\theta^{2k}}{2k!} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^{2k} + \sum_{k=0}^{\infty} \frac{\theta^{2k+1}}{(2k+1)!} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^{2k+1}$$

$$= \underbrace{\sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k}}{2k!}}_{\cos\theta} \mathbb{1} + \underbrace{\sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k+1}}{(2k+1)!}}_{\sin\theta} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

$$\exp\left[\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}\theta\right] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix}$$

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look at commutators: enough to check on basis

$$[Y_x, Y_y] = Y_z$$

$$[Y_y, Y_z] = Y_x$$

$$[Y_z, Y_x] = Y_y$$

<u>representations</u> of Lie algebra $\begin{cases} \text{irreducible} \\ \text{unitary} \\ \text{finite-dim} \end{cases}$

looking for a finite-dim complex vector space V
and an action of $\underline{so}(3)$ on V

$\pi : \underline{so}(3) \mapsto Lin(V, V)$ s.t. π linear

$[\pi(X), \pi(Y)] = \pi([X, Y])$

irreducibility: only <u>invariant</u> subspaces of V are $\{0\}$ and V

$\rightarrow$ $W \underset{\text{subspace}}{\subseteq} V$ invariant if $\pi(W) \subseteq W$

<u>trivial rep</u>: $\dim V = 1$ and $\pi(X) = 0 \quad \forall X \in \mathfrak{g}$

$\rightarrow$ irreducible

<u>Def</u>: $\pi : \mathfrak{g} \rightarrow Lin(V, V)$ unitary if $\pi(X)^\dagger = -\pi(X)$

<u>defining representation</u>: $\dim V = \dim \mathfrak{g}$

$\pi(X)$ is the linear map such that in a basis of V
X is the matrix rep. of $\pi(X)$

<u>adjoint rep</u> $V = \mathfrak{g}$ $\pi(X)(Y) = [X, Y]$

Heighest / lowest weight

- since V is complex, we can complexify the Lie algebra without breaking things

take $\mathfrak{g}$ map it to $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes \mathbb{C}$ (in other terms, we allow multiplication by complex scalars)

$\underline{so}(3) = \{ X \in M_3(\mathbb{R}) \mid X^t = -X \}$

$\underline{so}(3)_{\mathbb{C}} = \{ \alpha X \mid \alpha \in \mathbb{C},\ X \in \underline{so}(3) \} = \{ X \in M_3(\mathbb{C}) \mid X^t = -X \}$

$= \underline{sl}(3, \mathbb{C})$

since $\pi : \underline{so}(3) \to \mathrm{Lin}(V, V)$ is linear and $\pi(X)$ is complex-linear

we can extend π to $\underline{so}(3)_{\mathbb{C}}$

by saying if $X \in so(3)$, $\alpha \in \mathbb{C}$ $\quad \pi(\alpha X) = \alpha \pi(X)$

$J_x = i Y_x \qquad J_y = i Y_y \qquad J_z = i Y_z$

if unitary: $\pi(Y_z)^t = -\pi(z)$

$\pi(J_z)^t = \pi(J_z)$

$$\begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = e^{-iJ_z}$$

$J_\pm = J_x \pm i J_y$ $\qquad$ new basis $\{ J_z, J_+, J_- \}$

$[J_x, J_y] = i J_z \qquad [J_y, J_z] = i J_x \qquad [J_z, J_x] = i J_y$

$[J_+, J_-] = [J_x + i J_y, J_x - i J_y] = -i [J_x, J_y] + i [J_y, J_x]$

$= -2i [J_x, J_y] = 2 J_z$

$[J_+, J_-] = 2 J_z$

$[J_z, J_\pm] = \pm J_\pm$

$J_\pm$ ladder generators

$[J_z, J_x + i J_y] = [J_z, J_x] + i [J_z, J_y]$

$= i J_y + i(-i J_x) = J_x + i J_y$

$\pi(J_z)^t = \pi(J_z)$ $\qquad$ spectral theorem: $\pi(J_z)$ has a basis of eigenvectors

V has a basis of eigenvectors of J_z

if $|m\rangle$ is an eigenvector $J_z|m\rangle = m|m\rangle$ $m \in \mathbb{R}$

$[J_z, J_\pm] = \pm J_\pm$

$$J_z(J_\pm|m\rangle) = (J_\pm J_z + [J_z, J_\pm])|m\rangle$$

$$= m J_\pm|m\rangle \pm J_\pm|m\rangle = (m \pm 1) J_\pm|m\rangle$$

$J_\pm|m\rangle$ still an eigenvector of J_z with eigenvalue $m \pm 1$

must have m such that $J_+|m\rangle = 0$

$\pi(J_x)^\dagger = \pi(J_x)$ $\pi(J_y)^\dagger = \pi(J_y)$

$\rightarrow$ $\pi(J_\pm)^\dagger = \pi(J_\mp)$ $\begin{cases} \pi(J_+)^\dagger = \pi(J_-) \\ \pi(J_-)^\dagger = \pi(J_+) \end{cases}$

assume that $\langle m|m\rangle = 1$

<u>Casimir operator</u> $\pi(J_x)^2 + \pi(J_y)^2 + \pi(J_z)^2 \equiv J^2$

$$[J^2, \pi(J_i)] = 0$$

if $|m\rangle$ eigenvector of J_z then $J_z(J^2|m\rangle) = J^2 J_z|m\rangle$

$$= m(J^2|m\rangle)$$

$J^2|m\rangle$ eigenvector of J_z with same eigenvalue

$\rightarrow$ if an operator commutes with all the Lie algebra generators and rep is irreducible, then it acts as a multiple of identity